

Weekly Progress Report

LLM-Enhanced Dynamic Graph Networks for CVR Prediction

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7

Papers
Reviewed

2

Modules
Completed

1

GitHub
Repo

1

Overleaf
Doc

1. Literature Review

Module A — Delayed Feedback Correction (4 papers, completed)

DFM

Chapelle, KDD 2014

EM algorithm + exponential delay distribution to correct false-negative labels

ES-DFM

Yang et al., AAAI 2021

Elapsed-time sampling + importance weighting; curated data beats full noisy data

FSIW

Wen et al., 2021

Reframes as distribution shift; importance weights with consistency guarantee; 67x faster than DFM

DEFER

Su et al., arXiv:2104.14121

Duplicates positives AND real negatives to fix training distribution from the root; +8.5% CVR online

Module B — Graph Neural Networks for Recommendation (3 papers, completed)

LightGCN

He et al., SIGIR 2020

Pure neighbourhood aggregation without feature transform; ~16% gain over NGCF; shows ID embedding ceiling

TGN

Rossi et al., ICML 2020

Memory module (GRU) + Time2Vec for continuous-time dynamic graphs; state-of-the-art on link prediction

DGSR

Zhang et al., TKDE 2022

Dynamic graph for sequential rec; edge = (user, item, time, order); dual-channel attention (long + short term)

Module C — LLM-Based Recommendation (in progress this week: BERT4Rec, UniSRec, RLMRec, TALLRec)

2. Project Infrastructure

GitHub Repository

TripleLa-Liu/dissertation-cvr-llm

Private repo with structured README: architecture overview, dataset stats, baseline table, progress checklist

Overleaf Document

UCL Information Studies LaTeX Template

Title page, abstract, abbreviations list, and chapter skeleton (Ch.1–6) filled in with dissertation content

Google Doc

Shared with Dr. Arabella Sinclair

Literature notes, dataset comparison, meeting Q&A, and supervisor feedback all documented

Proposal (updated)

Proposal_Liu_Yize_Updated.docx

Revised proposal submitted via email: updated title, three RQs, three-layer methodology, and bibliography

3. Next Steps

This week

Complete LLM recommendation paper review (BERT4Rec, UniSRec, RLMRec, TALLRec)

Next week

Finalise dataset selection (Ali-CCP primary / Criteo supplementary); begin preprocessing

Week 3–4

Draft Methodology chapter; implement baseline models (DGSR, DEFER, LightGCN)